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Dr Carl Anthony, Head of the Department of Mechanical Engineering
School of Engineering
University of Birmingham
Edgbaston, Birmingham B15 2TT, United Kingdom

Dear Dr Anthony and Members of the Selection Committee,

Application for Assistant Professor (Research and Education) in Mechanical Engineering, Advanced Propulsion and Vehicle Systems, post 58193

I am writing to apply for the **Assistant Professor (Research and Education) in Mechanical Engineering** post (**58193**, Grade 8) in the **Department of Mechanical Engineering**, **School of Engineering**, at the University of Birmingham. I am currently a Research Assistant at the **Power Electronics and Machines Centre (PEMC), University of Nottingham**, having completed my **PhD in Electrical Engineering at Southeast University (SEU), Nanjing**, in September 2025 with the dissertation *Multi-physics Analysis and Design of Hairpin Winding PM Traction Machines in Electric Vehicles*. My doctoral and postdoctoral work is well-aligned with the research areas named in the post, in particular **integrated electrified and hybrid powertrains**, **powertrain decarbonisation strategies**, and the **AI-driven** component of **intelligent, connected and automated vehicle systems**, and I would be glad to contribute to the Department's research and teaching on advanced propulsion and vehicle systems and the UK's net-zero transport agenda.

My primary research focus is the **multi-physics design and thermal management of high power-density EV traction machines** for next-generation electrified powertrains. My *IEEE Trans. Transportation Electrification* paper *High-Fidelity Lumped Parameter Thermal Model for Oil-Cooling Hairpin Winding PM Machines in EVs* develops a high-resolution lumped-parameter network coupled to a fluid-solid oil-cooling model, validated against AVL dynamometer measurements. The same programme covers ***n*-layer flat-wire and hairpin winding design** for high power-density EV PM machines, **multi-objective optimisation** coupling electromagnetic, thermal and mechanical objectives, and **PWM-loss interaction analysis** that links converter behaviour to thermal-degradation limits. This is directly relevant to the post's interest in **integrated electrified and hybrid powertrains**, **thermal (HVAC) systems**, and **efficiency, degradation and lifecycle performance**.

I have also gained substantive **industrial development experience** through a multi-year collaboration with **Inovance**, a world-leading EV traction-machine supplier, on two oil-cooled PM traction machines, an **18 000 rpm 113 kW A0-class machine** (138 Nm, 7.78 kW/kg, 95.3% peak efficiency) and a **20 000 rpm 270 kW A-class machine** (380 Nm, 7.87 kW/kg, 97.4% peak efficiency). Each was taken through the full industrial cycle of multi-physics design, engineering-drawing release and tolerance specification, and AVL dynamometer characterisation across the full efficiency map. This combined academic and industrial background, together with my current PEMC role on the **Sustainable Multi-Sector Electrification Using Advanced Integrated Motor Drive Technologies (AIMD)** programme (Nottingham, Bristol and Newcastle), which covers both next-generation magnet-free synchronous machines and integrated motor-drive power electronics, should support competitive applications to **EPSRC**, **Innovate UK**, the **Advanced Propulsion Centre (APC)**, the **Faraday Institution** and the **Aerospace Technology Institute (ATI)**, as well as direct industrial co-funding from EV and powertrain partners.

A second research focus develops **AI-driven surrogate modelling and physics-grounded digital twins** for electric propulsion system design, addressing the post's interest in **intelligent, connected and automated vehicle systems empowered by AI**. My ICEM 2026 paper *A Deep Learning Based Magnetic Field Surrogate Modeling Method for Interior Permanent Magnet Machine in EVs* introduces an implicit physics-knowledge embedding approach in which a deep neural network learns to map machine design parameters to the **magnetic vector potential field**, reaching $\mathbf{R^2 = 0.9996}$ for average torque and **0.9730** for iron loss while reducing inference time by **more than 94%** against direct finite-element analysis. The surrogate sits on an **in-house**

Python finite-element code package I have developed (electromagnetic, structural-modal, mechanical-stress) and provides the substrate for **digital-twin** studies of propulsion components, fast design-space exploration, and AI-assisted control development. My methodological foundation in **multi-physics modelling of energy-conversion systems** extends naturally toward **hydrogen integration**, **fuel-flexible systems** and **whole-system efficiency optimisation**, and I would be glad to develop in those directions through interdisciplinary collaboration within the Department and with Birmingham's wider net-zero transport community.

My research record includes **22 journal papers, of which 17 are in IEEE Transactions** (across *IEEE Trans. Industrial Electronics*, *Transportation Electrification*, *Industry Applications*, and *Energy Conversion*), 18 peer-reviewed conference papers at IEEE IEMDC, IEEE ECCE, ICEM and ICEMS, and 9 patents. The work has been recognised by the **2025 IEEE TEC Electric Vehicle Design Competition (1st Prize)**, the **2025 IEEE Trans. Transportation Electrification Prize Paper Award (2nd Prize)**, the **1st Prize in the IEEE IAS CMD Thesis Contest (non-PhD category)**, the **CIEEC Best Paper Award (2023)**, and the **Jiangsu Province Outstanding Master Degree Thesis Award (2022)**. I also serve as **Co-Organiser and Co-Chair of an ICEM 2026 Special Session** on *Next-Generation Electrified Transport* in Funchal, Madeira, Portugal, and as peer reviewer for *IEEE Trans. Industry Applications*, *Energy Conversion*, and *Transportation Electrification*.

I would be glad to contribute to teaching across the Department's portfolio at undergraduate, MSc, CPD and short-course level, in areas such as **electric machines and drives, electrified powertrain design, multi-physics and thermal-management modelling, finite-element principles and FEA-based design, and AI for engineering**, drawing on the in-house Python finite-element code and the deep-learning surrogate as supporting material. During my doctoral studies I served as teaching assistant on the SEU course *Integrated Practice of Electrical Machine Systems*, preparing tutorial material, leading Q&A, reviewing project proposals, and contributing mark recommendations. I have also supported junior PhD and MSc colleagues on machine modelling, thermal-network construction and PWM-loss coupling.

My CV is attached, along with a detailed response to the selection criteria. I would be glad to discuss my research and teaching plans further at interview.

Yours sincerely,

Wentao Zhang